



VMWARE VSPHERE 7.0: INSTALL, CONFIGURE, MANAGE



Lab 09: Using Standard Switches

Document Version: 2020-11-04

Copyright © 2020 Network Development Group, Inc.
www.netdevgroup.com

NETLAB+ is a registered trademark of Network Development Group, Inc.

VMware is a registered trademark of VMware, Inc.

Contents

Introduction	3
Objectives.....	3
Lab Topology	4
Lab Settings	5
1 View the Standard Switch Configuration	6
2 Create a Standard Switch with a Virtual Machine Port Group	10
3 Attach Your Virtual Machines to the New Virtual Machine Port Group	15

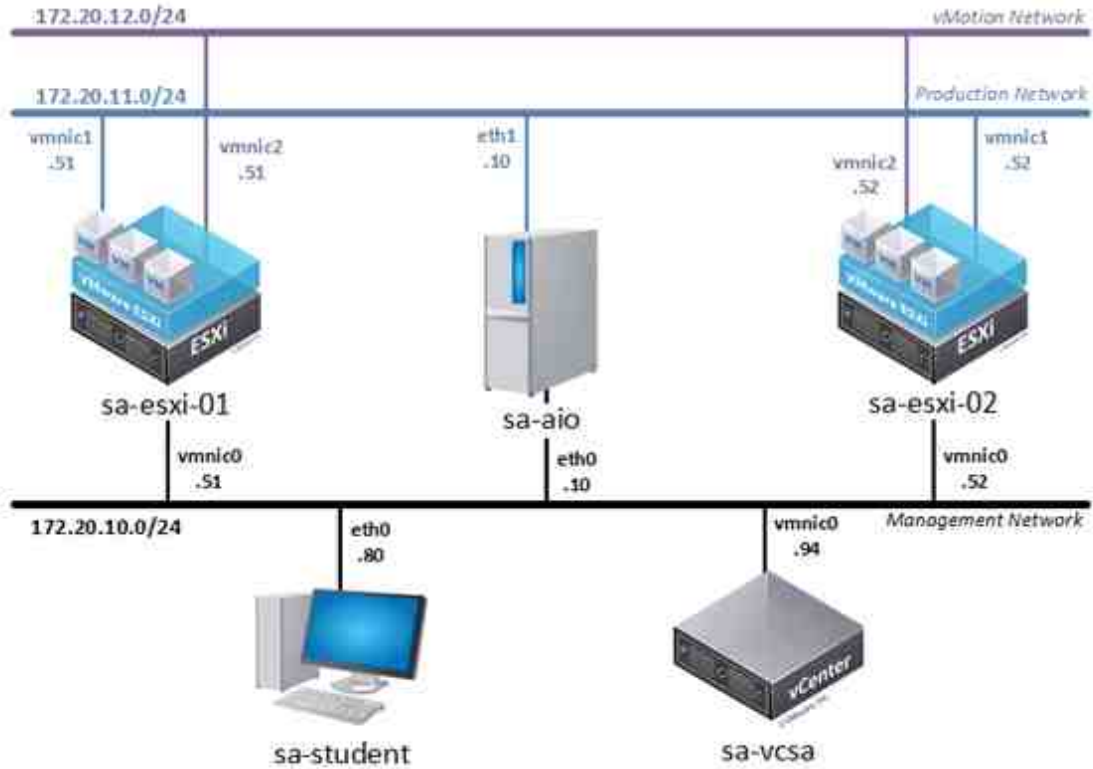
Introduction

In this lab, you will create a standard switch and port group.

Objectives

-) View the standard switch configuration
-) Create a standard switch with a virtual machine port group
-) Attach Virtual Machines to the virtual machine port group

Lab Topology



Lab Settings

The information in the table below will be needed in order to complete the lab. The task sections below provide details on the use of this information.

Virtual Machine	IP Address	Account	Password
sa-esxi-01	eth0: 172.20.10.51	root	Train1ng\$
sa-esxi-02	eth0: 172.20.10.52	root	Train1ng\$
sa-aio	eth0: 172.20.10.10	sysadmin	vmware123
sa-student	eth0: 172.20.10.80	sysadmin	vmware123
sa-vcasa	eth0: 172.20.10.94	sysadmin@vclass.local	vmware123

1 View the Standard Switch Configuration

In this task, you will view the VMware vSphere standard switch settings to ensure the proper configuration of the default switch.

1. Launch the sa-student virtual machine to access the graphical login screen.

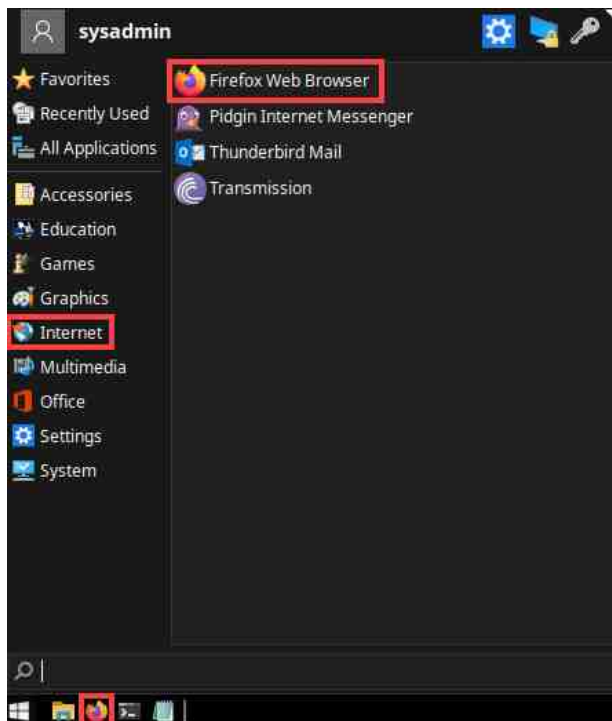


To launch the console window for a virtual machine, you may access by either clicking on the machine's graphic image from the topology page or by clicking on the machine's respective tab from the navigation bar.

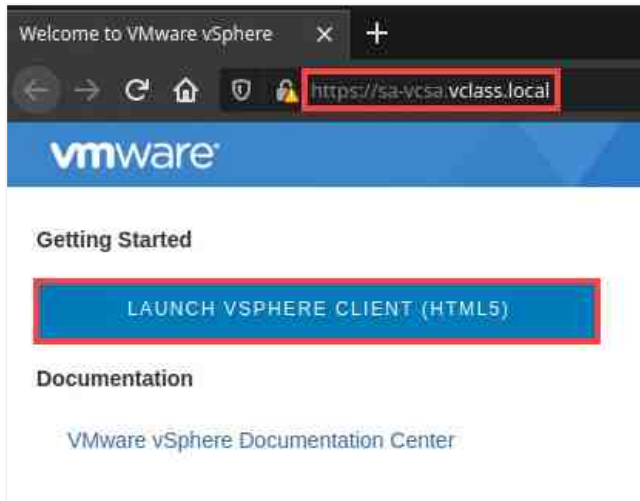
2. Log in as `sysadmin` using the password `vmware123`.



3. Launch the Mozilla Firefox web browser by either clicking on the icon shortcut found on the bottom toolbar or by navigating to Start Menu > Internet > Firefox Web Browser.

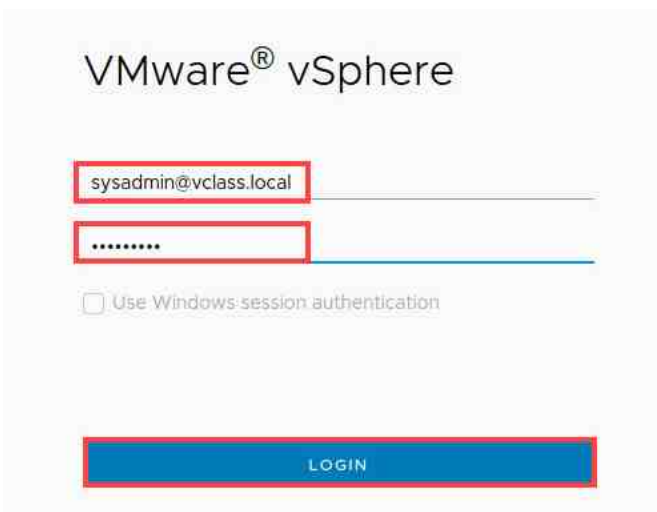


4. Notice the homepage is automatically configured to load the URL address of the VMware vCenter Server Appliance (sa-vcasa virtual machine). Click on the LAUNCH VSPHERE CLIENT (HTML5) button to launch the vSphere Client.



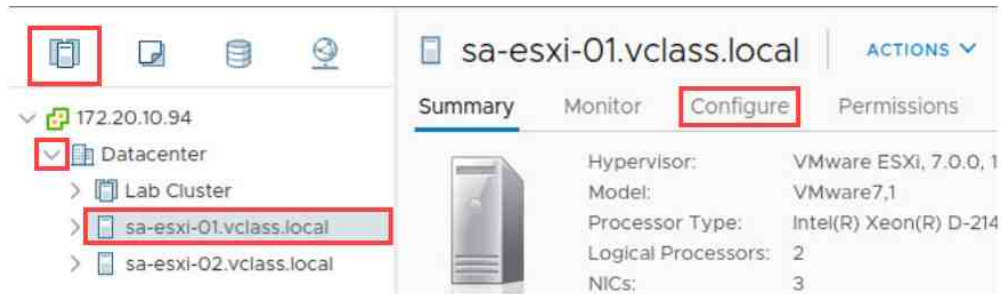
If you cannot successfully load the VMware Getting Started webpage, then please wait an additional 3-5 minutes and refresh the page to continue. During this time, the vCenter Server Appliance is still booting up and requires extra time to initialize.

5. In the Username text field, type `sysadmin@vclass.local` and in the Password text field, type `vmware123`. Click on LOGIN.

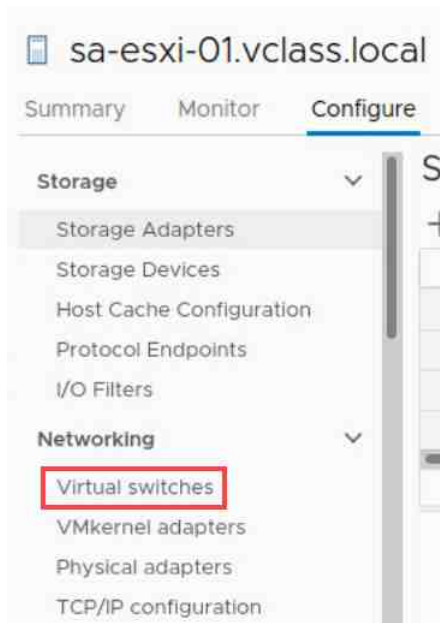


You may ignore the “browser-OS combination” warning message presented on the VMware vCenter Single Sign-On page and continue moving forward with the lab.

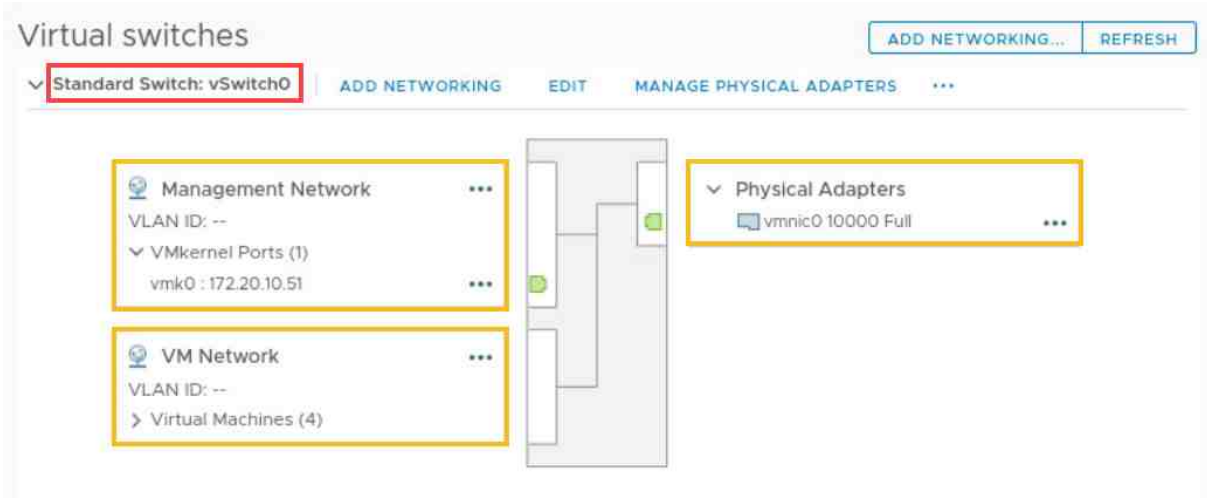
6. Ensure that you are on the Hosts and Clusters tab in the navigator pane. Expand the Datacenter inventory object, select sa-esxi-01.vclass.local, followed by clicking on the Configure tab in the main workspace.



7. In the main workspace, in the middle pane, select Virtual switches.



8. In the Virtual switches pane, ensure that Standard Switch: vSwitch0 is selected and review the information for the virtual switch. Notice the name for the default standard switch, the physical adapter that the standard switch is connected to, which network your virtual machines are connected to, along with which networks are connected to the default standard switch.

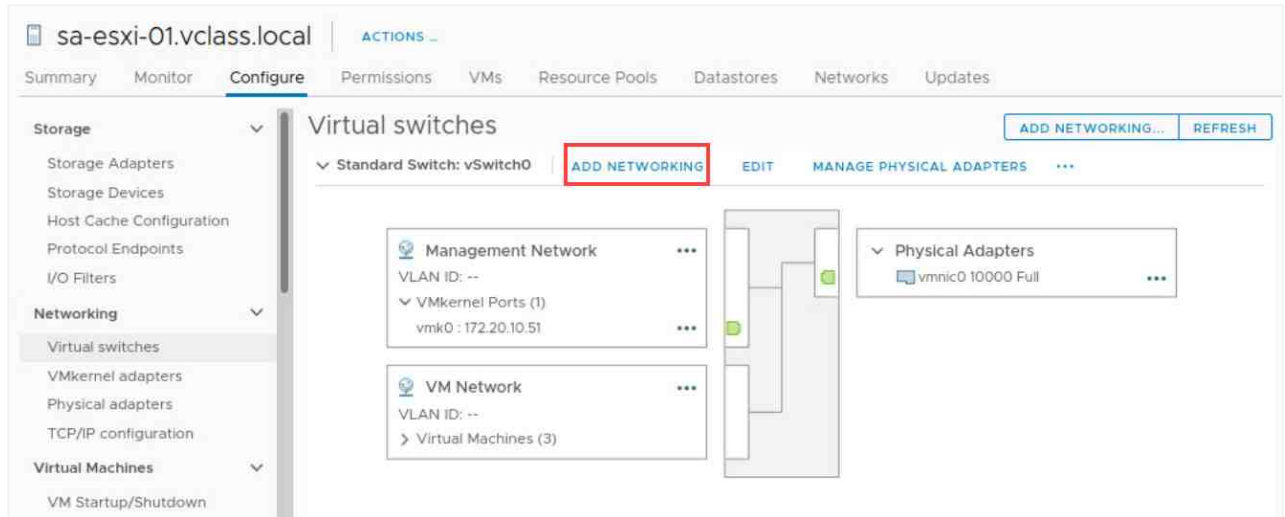


9. Leave the vSphere Client open and continue to the next task.

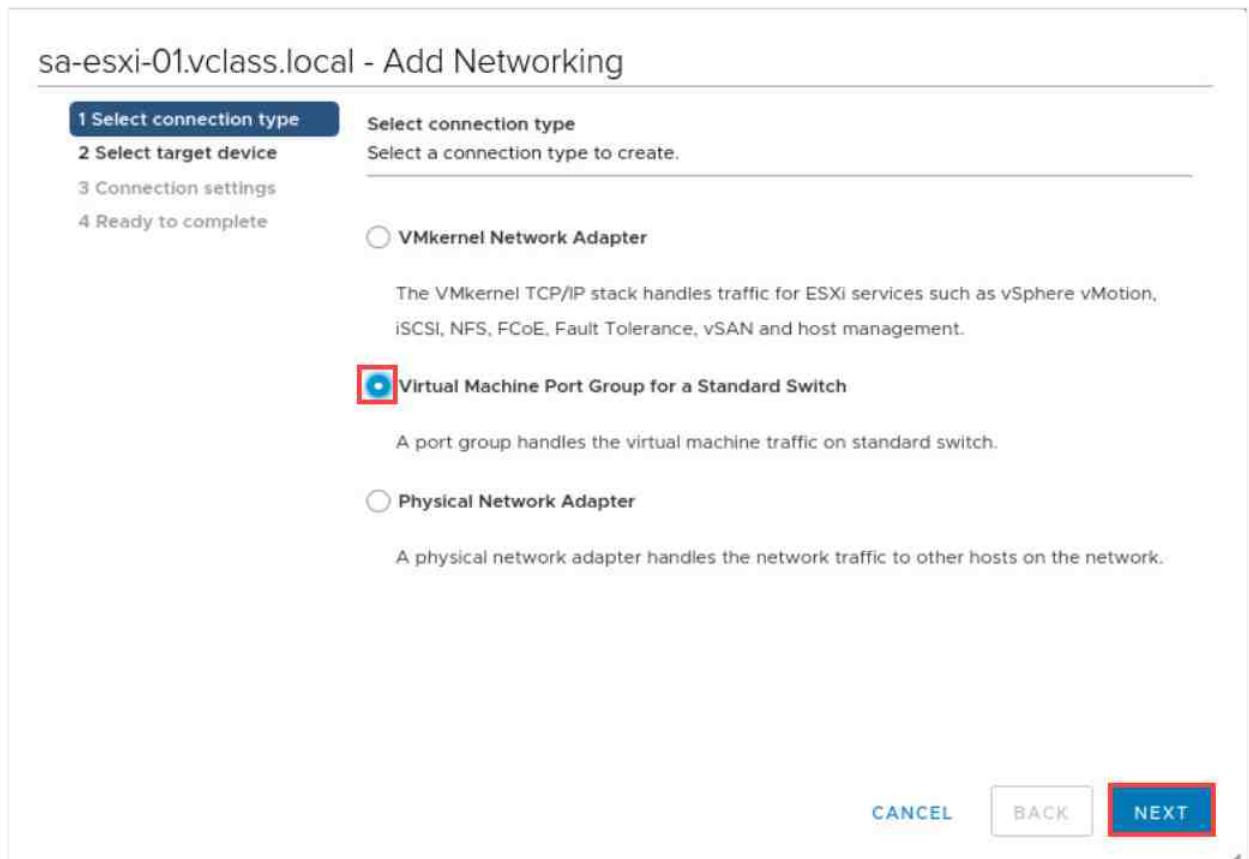
2 Create a Standard Switch with a Virtual Machine Port Group

In this task, you will create a virtual machine port group on a standard switch, which handles network traffic at the host level in your vSphere environment.

1. While in the vSphere Client, in the main workspace, click on Add Networking.



2. In the Add Networking window, on the Select connection type step, select the radio button for Virtual Machine Port Group for a Standard Switch and click Next.



- On the Select target device step, select the radio button for the New standard switch and click Next.

sa-esxi-01.vclass.local - Add Networking

✓ 1 Select connection type

2 Select target device

3 Create a Standard Switch

4 Connection settings

5 Ready to complete

Select target device

Select a target device for the new connection.

☐ Select an existing standard switch

vSwitch0 BROWSE ...

☒ **New standard switch**

MTU (Bytes) 1500

CANCEL BACK **NEXT**

- On the Create a Standard Switch step, click the Add adapters icon (the green plus icon).

sa-esxi-01.vclass.local - Add Networking

✓ 1 Select connection type

✓ 2 Select target device

3 Create a Standard Switch

4 Connection settings

5 Ready to complete

Create a Standard Switch

Assign free physical network adapters to the new switch.

Assigned adapters

☒ ☐ ☐ ☐

Active adapters

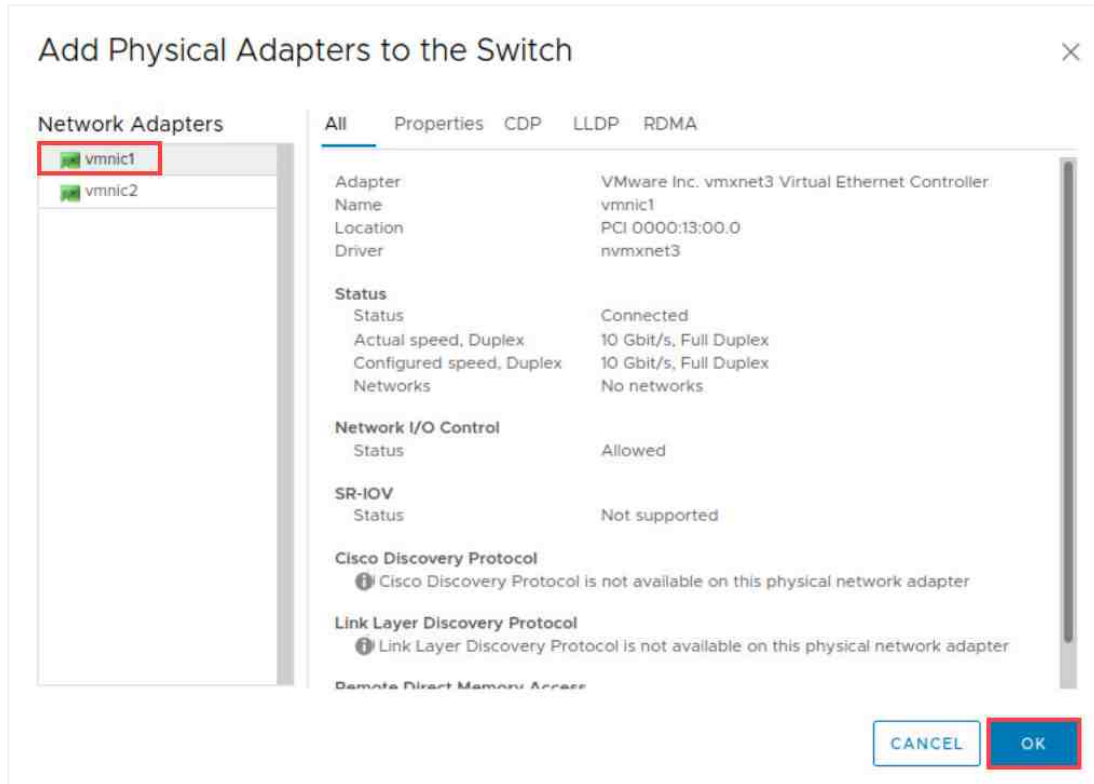
Standby adapters

Unused adapters

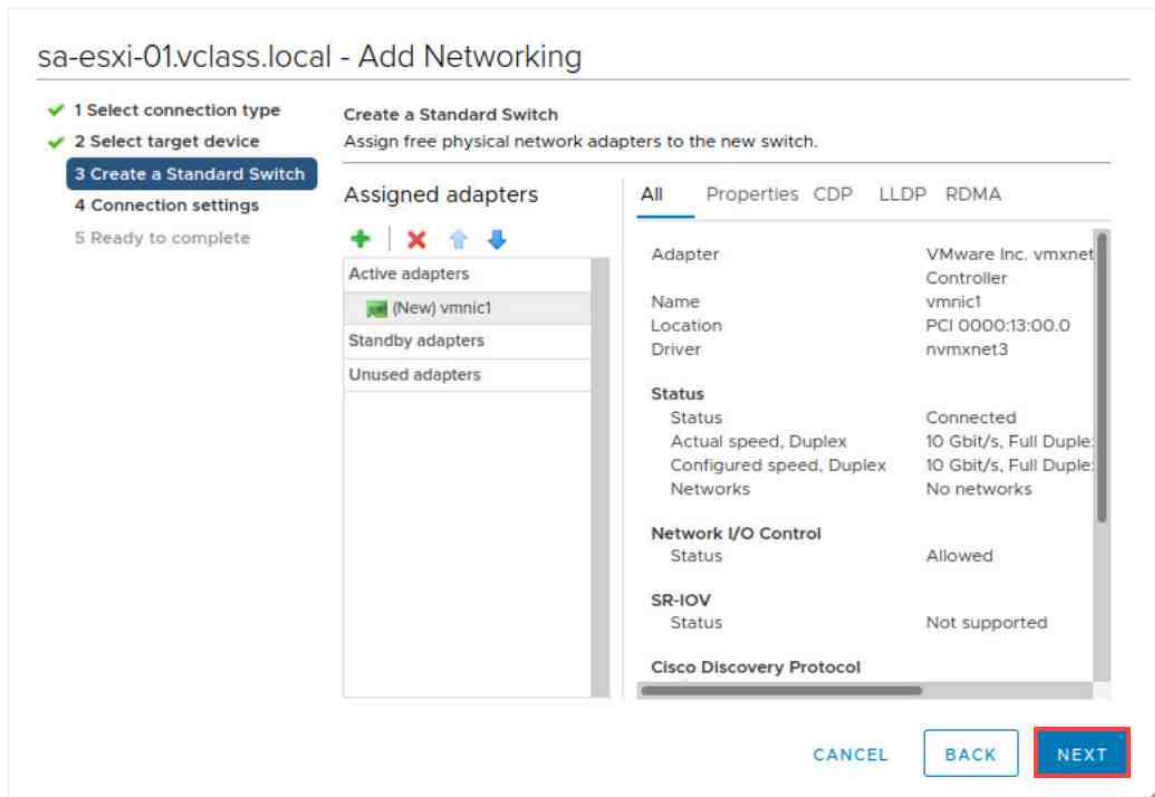
Select a physical network adapter from the list to view its details.

CANCEL BACK **NEXT**

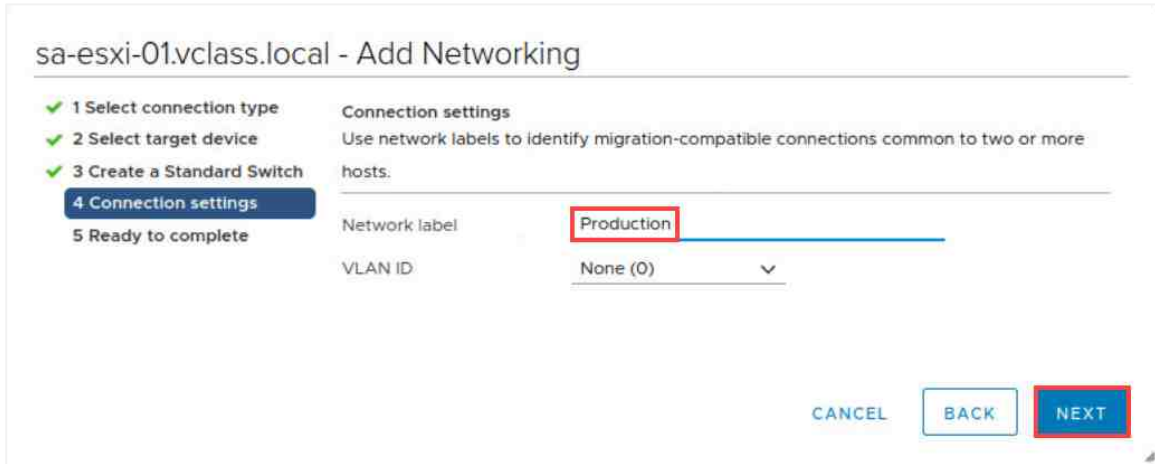
- In the Add Physical Adapters to the Switch window, select vmnic1 for the virtual switch and click OK.



- Review the information for the new active adapter and click Next.



- On the Connection settings step, enter **Production** in the Network label text box and click **Next**.



sa-esxi-01.vclass.local - Add Networking

1 Select connection type
2 Select target device
3 Create a Standard Switch
4 Connection settings
5 Ready to complete

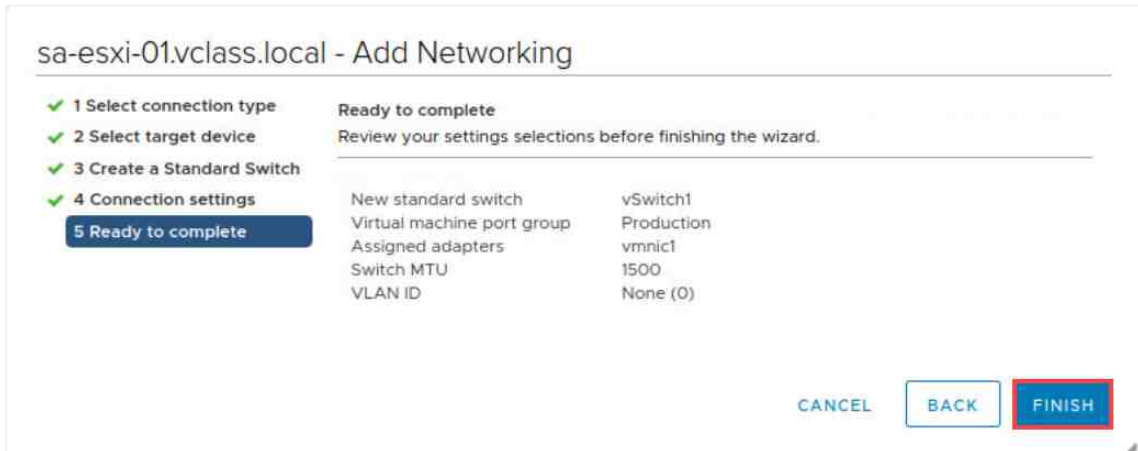
Connection settings
Use network labels to identify migration-compatible connections common to two or more hosts.

Network label: **Production**

VLAN ID: None (0)

CANCEL BACK NEXT

- On the Ready to complete step, verify that the information is accurate and click **Finish**.



sa-esxi-01.vclass.local - Add Networking

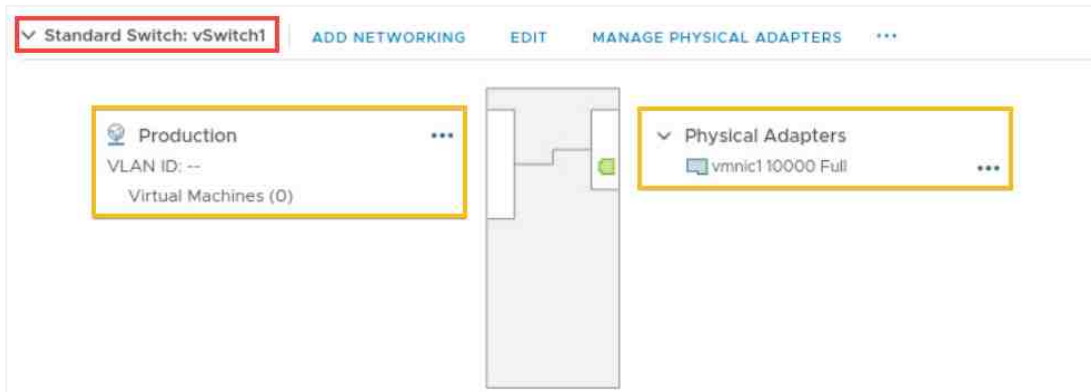
1 Select connection type
2 Select target device
3 Create a Standard Switch
4 Connection settings
5 Ready to complete

Ready to complete
Review your settings selections before finishing the wizard.

New standard switch	vSwitch1
Virtual machine port group	Production
Assigned adapters	vmnic1
Switch MTU	1500
VLAN ID	None (0)

CANCEL BACK FINISH

- In the Virtual switches pane, click on **Standard Switch: vSwitch1** to expand the view. Notice the **Production** group appears, and it has the Physical Adapter of **vmnic1**.



Standard Switch: vSwitch1 ADD NETWORKING EDIT MANAGE PHYSICAL ADAPTERS ...

Production
VLAN ID: --
Virtual Machines (0)

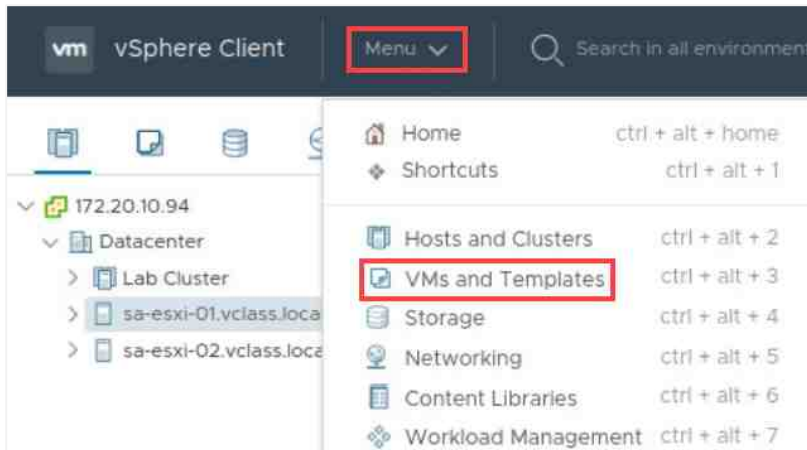
Physical Adapters
vmnic1 10000 Full

10. Repeat Task 2 for your second ESXi host (sa-esxi-02.vclass.local).
11. Leave the vSphere Client open to continue with the next task.

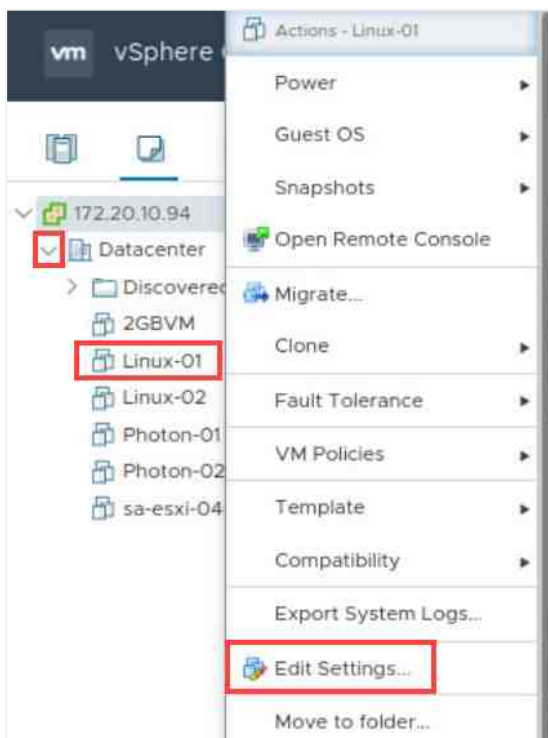
3 Attach Your Virtual Machines to the New Virtual Machine Port Group

In this task, you will attach two virtual machines to a virtual switch port group so that the virtual machines can communicate with other networked devices.

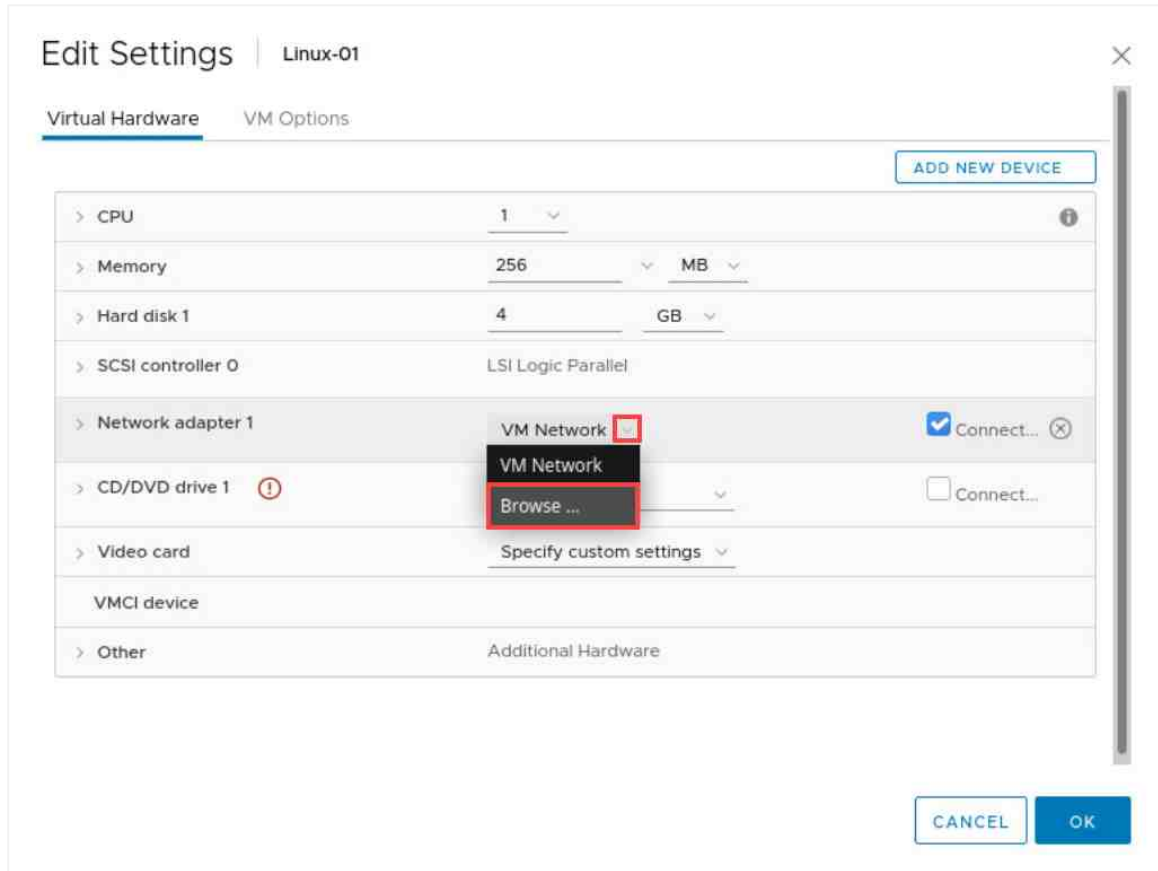
1. While in the vSphere Client, click the Menu and select VMs and Templates.



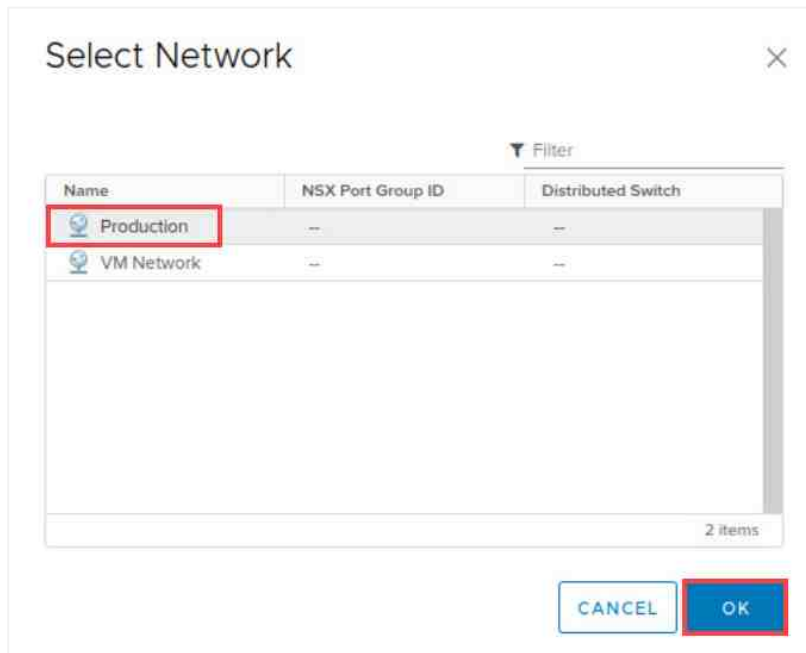
2. In the navigator pane, expand the Datacenter object. Right-click Linux-01 and select Edit Settings.



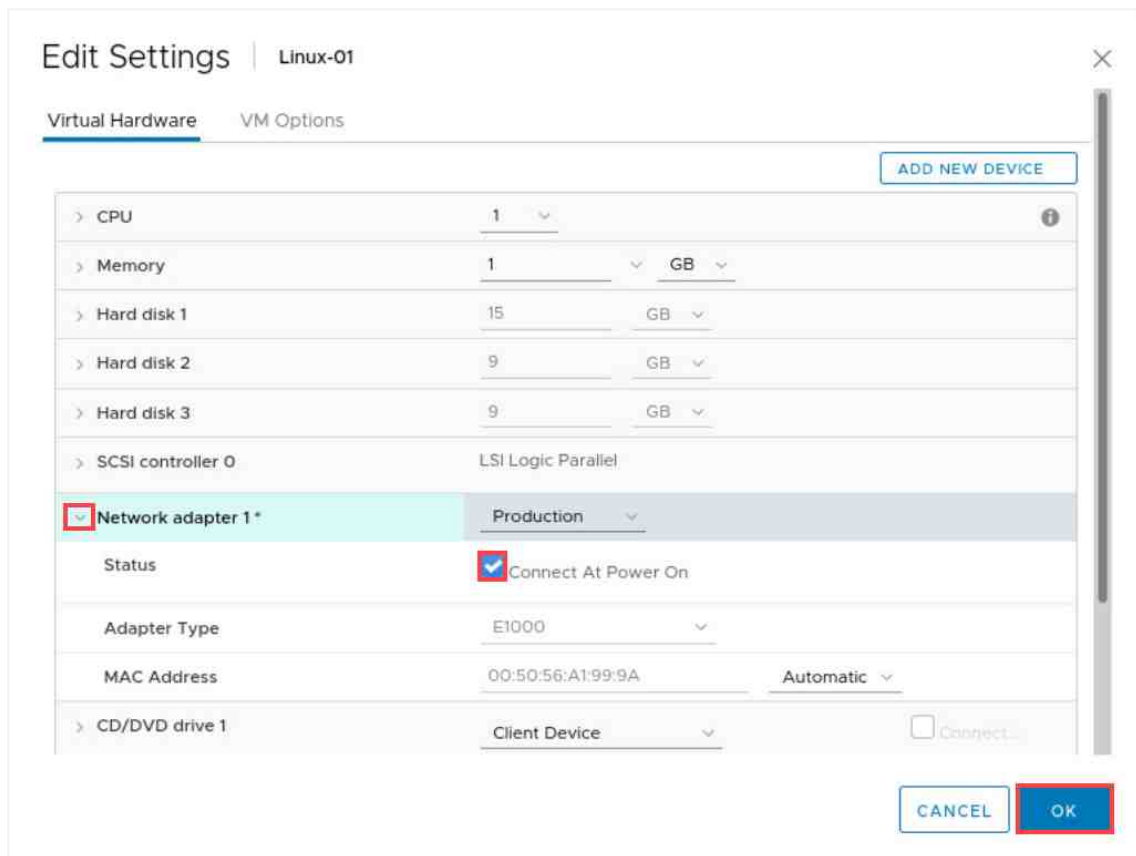
3. In the Edit Settings window, click the dropdown menu for the Network adapter 1 and select Browse.



4. In the Select Network window, select the Production network and click OK.



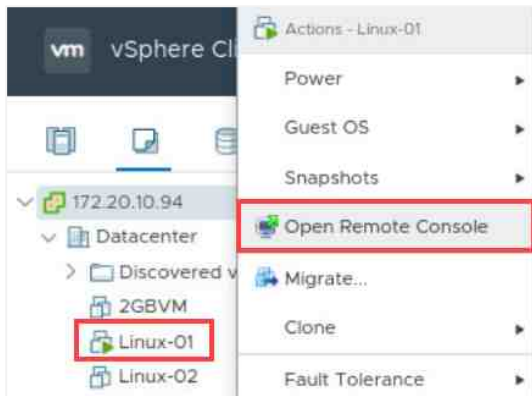
- Expand the view for Network adapter 1 by clicking the arrow and verify that the Connect At Power On checkbox is selected. Click OK to save the configurations.



- Power on the machine by right-clicking on Linux-01 from the navigator pane and selecting Power > Power On.



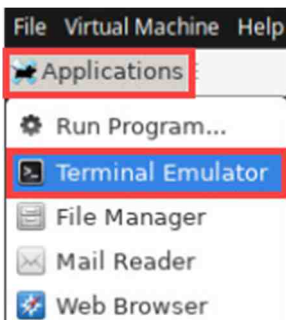
7. Right-click on Linux-01 and select Open Remote Console.



8. Log in to the virtual machine as sysadmin with the password of vmware123.



9. Open the terminal by selecting Applications and clicking on Terminal Emulator.



10. At the prompt, enter the command below and verify the IP address is set to 172.20.11.21.

```
Linux-01:~$ ifconfig
```

```
Linux-01:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:50:56:A1:EE:51
          inet addr:172.20.11.21  Bcast:0.0.0.0  Mask:255.255.255.0
          inet6 addr: fe80::250:56ff:fe01:ee51/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:26 errors:0 dropped:0 overruns:0 frame:0
          TX packets:39 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1800 (1.7 KiB)  TX bytes:2836 (2.7 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

11. Ping the default gateway IP address of 172.20.11.10 (sa-aio) by entering the command below at the command prompt.

```
Linux-01:~$ ping -c4 172.20.11.10
```

```
Linux-01:~$ ping -c4 172.20.11.10
PING 172.20.11.10 (172.20.11.10): 56 data bytes
64 bytes from 172.20.11.10: seq=0 ttl=42 time=0.834 ms
64 bytes from 172.20.11.10: seq=1 ttl=42 time=0.530 ms
64 bytes from 172.20.11.10: seq=2 ttl=42 time=0.469 ms
64 bytes from 172.20.11.10: seq=3 ttl=42 time=0.615 ms

--- 172.20.11.10 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 0.469/0.612/0.834 ms
Linux-01:~$
```

12. Notice the successful pings where Linux-01 can now communicate with sa-AIO. Repeat Task 3 for the Linux-02 virtual machine and verify the IP address is 172.20.11.22. Once you have verified the IP address, ping the default gateway.
13. As an extra step, confirm that both Linux-01 and Linux-02 virtual machines can communicate with each other on the Production network by using the ping command.
14. The lab is now complete; you may end the reservation.